

HEALTHCARE DATA ANALYTICS

USING POWER BI

PROJECT REPORT

Project Title

Healthcare Data Analytics Dashboard

Submitted By

SURAJ SHARMA

Course

Data Analytics

Tools Used

Power BI Desktop • Power Query • DAX

1. Project Objective

The objective of this project is to analyze healthcare data using Power BI and identify insights related to patient demographics, health conditions, hospital visits and treatment costs. An interactive dashboard was created to help stakeholders understand the healthcare data.

2. Dataset Description

The dataset contains 500 patient records and 10 columns.

Column	Description
Patient_ID	Unique patient identifier
Age	Age of patient
Gender	Gender of patient
BMI	Body Mass Index
Blood_Pressure	Patient blood pressure
Cholesterol_Level	Cholesterol level
Diabetes	Diabetes status

Smoker	Smoking status
Hospital_Visits	Number of hospital visits
Treatment_Cost	Cost of treatment

3. Data Preparation

The dataset was imported into Power BI and cleaned using Power Query. Column names and data types were checked, missing values were reviewed, and the table was renamed to **Healthcare_Data**. The cleaned data was then loaded into Power BI.

4. Data Model

The project contains one table named **Healthcare_Data**. Since there is only one table, no relationship between tables is required.

Healthcare_Data: Patient_ID, Age, Gender, BMI, Blood_Pressure, Cholesterol_Level, Diabetes, Smoker, Hospital_Visits, Treatment_Cost

5. DAX Calculations

Average Treatment Cost

```
Average Treatment Cost =
AVERAGE(Healthcare_Data[Treatment_Cost])
```

Average BMI

```
Average BMI =
AVERAGE(Healthcare_Data[BMI])
```

Total Hospital Visits

```
Total Hospital Visits =
SUM(Healthcare_Data[Hospital_Visits])
```

Age Group

```
Age Group =
SWITCH(
TRUE(),
Healthcare_Data[Age] <= 30, "18-30",
Healthcare_Data[Age] <= 45, "31-45",
Healthcare_Data[Age] <= 60, "46-60",
"61+"
)
```

6. Dashboard Components

- Average Treatment Cost – KPI
- Average BMI – KPI
- Total Hospital Visits – KPI
- Age Distribution of Patients
- Gender Distribution of Patients
- BMI vs Treatment Cost scatter plot
- Average Hospital Visits by Age Group
- Smoking Status pie chart
- Diabetes Status pie chart
- Gender, Diabetes and Smoking filters

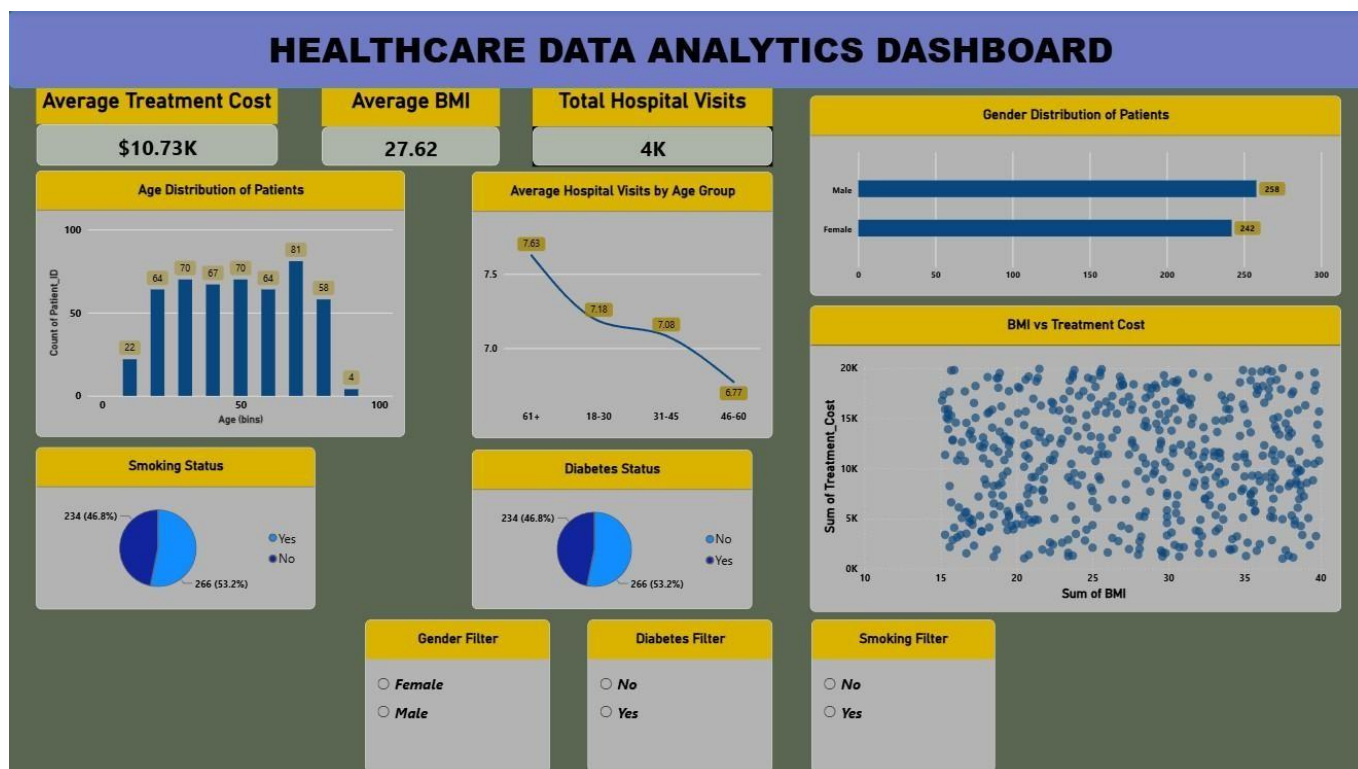


Figure 1: Final Healthcare Data Analytics Dashboard

7. Dashboard Interpretation

The dashboard provides a visual summary of patient demographics, hospital visits, treatment costs, BMI, diabetes status and smoking status. The slicers allow users to filter the dashboard by Gender, Diabetes and Smoker.

8. Key Insights

Analysis	Result / Insight
Age group with highest average hospital visits	61+ has the highest average hospital visits at approximately 7.63 visits. The other groups are
Diabetes vs treatment cost	Non-diabetic patients have an average treatment cost of approximately 11,021.85, while
BMI vs treatment cost	The scatter plot shows no meaningful linear relationship between BMI and treatment cost
Gender distribution	There are 258 male patients and 242 female patients.
Smoking status	Average hospital visits are very similar: approximately 7.28 for non-smokers and 7.24 for

Interpretation note: These are descriptive observations from this dataset and should not be interpreted as proof that diabetes or smoking causes treatment costs or hospital visits to change.

9. Conclusion

The Healthcare Data Analytics project demonstrates how Power BI can be used to analyze patient demographics, health conditions, hospital visits and treatment costs. The interactive dashboard provides a clear visual summary and allows filtering by gender, diabetes and smoking status.

10. Tools & Technologies

- Power BI Desktop
- Power Query

- DAX
- CSV Dataset
- Data Visualization

11. Project Deliverables

- Power BI (.pbix) file
- PDF project report
- Dashboard screenshot
- Data model documentation
- DAX documentation
- Key insights and interpretation

12. Final Dashboard Checklist

- ✓ Dashboard title is present.
- ✓ Three KPI cards are present.
- ✓ Age distribution chart is present.
- ✓ Gender distribution chart is present.
- ✓ BMI vs Treatment Cost scatter plot is present.
- ✓ Average Hospital Visits by Age Group chart is present.
- ✓ Smoking and Diabetes pie charts are present.
- ✓ Gender, Diabetes and Smoking slicers are present and functional.
- ✓ Dashboard screenshot is included in this report.